

		Marks	Credits
L T P	Exam:	100	4
4 - -	Sessional:	50	
	Total:	150	4

Duration of Exam: 3 hrs.

Course Outcomes:

Upon completing the course, the student will:

CO1 Be familiar with the basics concepts in theory of computation;

CO2 Be able to construct finite state machines and the equivalent regular expressions.

CO3 Be able to construct pushdown automata and their equivalent context free grammars.

CO4 Be exposed to the advanced concepts of theory of automata computation.

NOTE: Examiner will set nine question in total. Question One will be compulsory and will comprise of all section and remaining eight questions to be set by taking two questions from each unit. The students have to attempt five questions in total, first being compulsory and selecting one from each Unit.

UNIT 1

Regular Languages: Finite automata, DFA, NFA, Equivalence of DFA & NFA. An application, Mealy and Moore Models, Regular expressions and languages. Context free languages: CFGs, Applications, Ambiguity removal, CNF, GNF.

UNIT 2

PushDown Automata: Basics of PDA, Acceptance By PDA, PDA and CFL, Parsing and PDA: Top Down Parsing and Bottom up Parsing

UNIT 3

Turing Machine: Turing machines, variants of TMs, Restricted TMs, TMs and Computers.**Decidability:** Decidable languages, decidable problems concerning Context free languages, the halting problem, halting problem is undecidable.

UNIT 4

Reducibility and Computability: Undecidable problems from language theory – Regular expressions, Turing machines, Reduction. A simple undecidable problem (PCP), Primitive recursive functions, tractable decision problems, theory of Optimization, Church- Turing Thesis.

References:

1. Introduction to Theory of Computation – Michael Sipser (Thomson Nrools/Cole)
2. Introduction to Automata Theory, Languages and Computations – J.E. Hopcroft, Rajeev Motwani& J.D. Ullman (Pearson Education Asia), 2nd Edition.
3. Theory of Computation by Peter Linz
4. Introduction to languages and theory of computation – John C. Martin (MGH)